

Wage Inequality and the Spatial Expansions of Firms

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Multi-region firms: a growing trend

- Firms with establishments across multiple markets are more and more common
- This is very relevant in services: food, finance, healthcare..

Year	Firms	% of workforce	Emp. per firm	Estabs. per firm	CZs per firm
1980	43500	24%	432.6	12.2	5.0
2017	72500	41%	748.2	20.4	6.4

This paper argues this spatial expansion has played a relevant role in labor markets in the U.S. and can account for a big part of the rise in inequality in many dimensions:

- Across establishments
- Across firms
- Across space

- Empirical Evidence

- Document stylized facts that guide the model
- Key findings:
 - Spatial expansion in the past 40 years has been remarkable
 - HQ activity is concentrated spatially and skill-intensive
 - Multi-region service firms account for a significant part of the rise in inequality

- Theory:

- Develop a general equilibrium model with services firms operating branches
- Generate implications for inequality within workers and firms

- Estimation:

- Estimate the model for 391 labor market areas with an SSM approach
- Perform counterfactual analysis by shocking the 1980 equilibrium
- Find the model can explain secular trends in US labor markets

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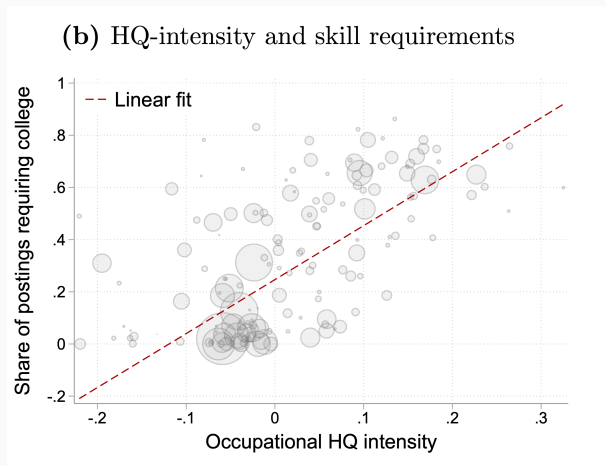
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Empirical evidence

- Main data source: U.S. Census Bureau Longitudinal Business Database (LBD)
 - Establishment-level data on employment, payroll, location, NAICS code
 - Industry and sector defined with NAICS code that accounts for largest share of payroll
 - A firm is in services if at least half of payroll is (according to BEA)
- 391 Labor market areas (LMAs) are aggregations of Tolbert and Sizer (1996) CZs
- Data augmented to get job level or worker-level information
 - Burning glass technologies (BGT): info about spatial distribution of jobs
 - Dun & Breadstreet business files to identify HQ from BGT and help estimation

Firm spatial structure

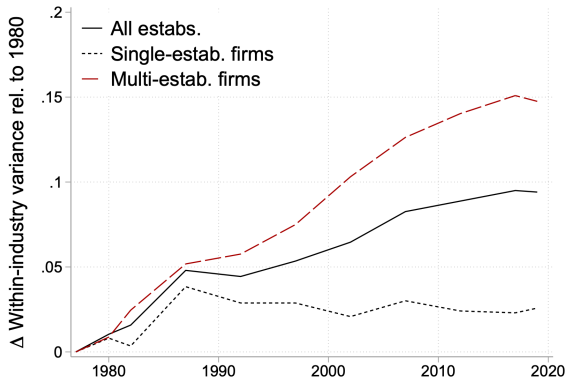
- Headquarters activity is spatially concentrated within firms [Details](#)
- And it has greater skill intensity, higher wages than branch-level activity: [Wage premium](#)



Multi-region firms and wage inequality

- Multi-region firms account for most of the rise in within-industry wage inequality

(b) Δ Variance within industries by firm type



Can be further decomposed to show that:

- Due both to reallocation to these firms and rising inequality for this group of firms
- 45 % of the rise of inequality in multi-region service firms is between establishments within firms

[Details](#)

Four main empirical facts:

1. Between 1980-2017, the average number of markets per firm and the share of employment in multi-region firms have both increased in the services-producing sectors
2. Headquarters activity is spatially concentrated within firms and it displays greater skill intensity and higher wages than branch-level activity
3. Multi-region firms account for most of the rise in within-industry wage inequality, both due to reallocation to these firms and rising inequality for this group of firms
4. Close to half of the rise in inequality for multi-region service firms is between different establishments within these firms

A spatial model of firm expansion

- N regions, S types of workers who differ in skill s , with supply L_s
- Households consume services and tradable-goods, choose their region i and employer ν (idiosyncratic shock)
- Tradable goods: homogeneous, freely traded and produced CRS
- Services: continuum of varieties with monopolistic competition
- Idiosyncratic preference shocks influence sorting decisions

Households

Households choose their region i , employer ν , consumption of tradable-goods Q_i^g and the bundle of local services Q_i to maximize

$$U_{i\nu} = b_{i\nu} (Q_i)^\beta (Q_i^g)^{1-\beta}, \quad Q_i = \left[\int_{\omega \in \Omega_i} (q_i(\omega))^{\frac{\sigma-1}{\sigma}} d\omega \right]^{\frac{\sigma}{\sigma-1}}$$

- Price indexes $P_i = \left(\int_{\omega \in \Omega_i} P_i(\omega)^{1-\sigma} d\omega \right)^{\frac{1}{1-\sigma}}$ and $\bar{P}_i = P_i^\beta$.
- $b_{i\nu}$ is an household-specific idiosyncratic preference shock for region i , employer ν
 - Drawn from a nested Fréchet distribution $F(\mathbf{b}) = \exp \left(- \sum_{i=1}^N \left(\int_{\nu \in \mathcal{V}_i} b_{i\nu}^{-\epsilon} d\nu \right)^{\frac{\xi}{\epsilon}} \right)$
 - Lower nest parameter ϵ captures employer-level labor supply elasticity
 - Upper nest parameter ξ captures regional labor supply elasticity

Upward sloping labor supply

- The regional labor supply:

$$\frac{L_{is}}{\sum_{j=1}^N L_{js}} = \frac{(W_{is}/\bar{P}_i)^\xi}{\sum_{j=1}^N (W_{js}/\bar{P}_j)^\xi}$$

- Conditional on choosing i , the mass of type- s workers that choose employer ν is

$$l_{is}(\nu) = L_{is} \left(\frac{w_{is}(\nu)}{W_{is}} \right)^\epsilon$$

- So an employer ν in region i that hires $l_{is}(\nu)$ workers of skill s needs to pay them

$$w_{is}(\nu) = L_{is}^{-\frac{1}{\epsilon}} W_{is} l_{is}(\nu)^{\frac{1}{\epsilon}}, \quad W_{is} = \left(\int_{\nu \in \mathcal{V}_i} w_{is}(\nu)^\epsilon d\nu \right)^{\frac{1}{\epsilon}}$$

where W_{is} is the regional wage index for type- s workers.

The services production function

- Firm produces a unique variety ω , has HQ in i can open a branch in j by paying f
- A branch that hires l_{ijs} workers of type s produces $q_{ij}(\omega)$ units in market j according to

$$q_{ij}(\omega) = \varphi_{ij}(\omega) \prod_{s=1}^S l_{ijs}^{\alpha_s}$$

where

$$\varphi_{ij}(\omega) = A_{ij} z_{\omega,i} \phi_{\omega,j} \left(\prod_{s=1}^S h_{is}^{\rho_s} \right)^{\gamma}$$

- A_{ij} Exogenous bilateral shifter, common to all firms with HQ at i and a branch at j
- $z_{\omega,i}$ Firm-specific draw, common to all the firm's branches if it is HQed at i
- $\phi_{\omega,j}$ Firm-branch-specific draw in location j - no direct effect on other branches
- $(\prod_{s=1}^S h_{is}^{\rho_s})^{\gamma}$ Endogenous choice of HQ labor, affecting all the firm's branches

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Firm's production problem

Firm's problem: choose branch-level labor $\mathbf{l} \equiv \{l_{js}\}$ across all markets j and HQ-level labor $\mathbf{h} = \{h_{is}\}$ to maximize profits

$$\begin{aligned} \pi_i(z_i, \phi) = & \max_{h, l} \sum_j \underbrace{E_j^{\frac{1}{\sigma}} P_j^{\frac{\sigma-1}{\sigma}} q_{ij}(z_i, \phi)^{\frac{\sigma-1}{\sigma}}}_{\text{Sales in market } j} \\ & - \sum_j \mathbb{I}_j \underbrace{\left(\sum_s W_{js} L_{js}^{-\frac{1}{\epsilon}} l_{js}^{\frac{\epsilon+1}{\epsilon}} - \bar{p}_j f \right)}_{\text{Costs in market } j} - \underbrace{\sum_s W_{is} L_{is}^{-\frac{1}{\epsilon}} h_{is}^{\frac{\epsilon+1}{\epsilon}}}_{\text{HQ costs in market } i} \end{aligned}$$

where $\mathbb{I}_j$ is an indicator that equals to 1 if the firm is active in market j .

- Two additional assumptions:
 - Firms choose $\mathbf{h}$ before ϕ are observed, and $\mathbf{l}$ after
 - ϕ are drawn independently from a common Pareto distribution with parameter θ

Entry and headquarter location decisions

- Firms can enter freely in any initial market by paying f_e .
- Upon entry, firms observe their firm-level productivities $\mathbf{z}$ across all potential markets and choose where to locate their headquarters to solve

$$\max_i E_{\phi} [\pi_i (z_i, \phi)]$$

- Baseline productivities $z_{0,i}$ drawn from an i.i.d Fréchet with parameter $\eta > 1$.
- Introduce a simple friction: $z_i = cz_{0,i}$ if $i \neq n$, while $z_i = z_{0,i}$ if $i = n$
 - As $c \rightarrow 1$, the entry cost is sunk, and firms sort independently of their initial entry location.
 - As $c \rightarrow 0$, no ex-post sorting, and HQ location=initial entry location

Equilibrium characterization: firm wages and inequality

The wages that a firm pays for workers of type s admit a log-linear structure given by

$$\log w_{ijs}(z, \phi) = \text{const}_s + \underbrace{\frac{1}{\epsilon+1} \log \left(W_{js}^\epsilon L_{js}^{-1} \bar{p}_j^{\frac{\theta-1}{\theta}} \Upsilon_j^{\frac{1}{\theta}} \right)}_{\text{Local market effect}} + \underbrace{\frac{\chi}{\epsilon} \log \phi}_{\text{Branch-specific shock}}$$

$$+ \underbrace{\frac{\chi}{\epsilon} \frac{1}{1 - \gamma\chi\theta} \log z}_{\text{Fundamental firm productivity}} + \underbrace{\frac{\gamma\chi}{\epsilon+1} \log \frac{\Lambda_i}{\Gamma_i}}_{\text{HQ Market effect}} + \underbrace{\frac{\chi}{\epsilon} \log A_{ij}}_{\text{HQ-branch frictions}}$$

with $\chi \equiv \frac{\frac{\epsilon}{\epsilon+1} \frac{\sigma-1}{\sigma}}{1 - \frac{\epsilon}{\epsilon+1} \frac{\sigma-1}{\sigma}}$ and regional aggregates:

- Υ_j : market-potential of operating a branch in j
- Λ_i : attractiveness of operating headquarters in i
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Equilibrium characterization: inequality across firms and workers

- Consider two firms with HQ in i , with productivities $z'_i > z_i$, active in j with same ϕ_j . Then

$$\frac{w_{is}^{HQ}(z')}{w_{ijs}(z', \phi)} > \frac{w_{is}^{HQ}(z)}{w_{ijs}(z, \phi)}$$

Details

For firm type

- The economy-wide payroll-share of high-skilled workers is independent of the spatial distribution of economic activity, and it's increasing: Details
- in the share of services β when services are more skill-intensive than tradables:
$$\frac{\gamma}{\gamma+1}\rho_s + \frac{1}{\gamma+1}\alpha_s > \alpha_{g,s}$$
- in the HQ-branch elasticity γ when HQs are more skill-intensive than branches: $\rho_s > \alpha_s$

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Inequality across worker types and spatial segregation

- Location i 's specialization in providing headquarters can be measured as

$$\frac{\text{Total sales by locally HQed firms}}{\text{Domestic expenditure on services}} = \frac{R_i}{E_i}$$

- The payroll share of high-skilled workers in region i is increasing in R_i/E_i if:
 - Headquarters are sufficiently skill intensive: $\rho_s > \alpha_{g,s} + \frac{1}{\gamma} (\alpha_{g,s} - \alpha_s)$
 - The productivity loss from reallocation of headquarters is small: ($c \rightarrow 0$)
- In the limit economy with no multi-region service firms, i.e. $A_{ij} \rightarrow 0 \forall i \neq j$, there are no differences in the payroll-share of high-skilled workers across space.

Model Quantification

Setting up estimation

We wish to quantify the effect of firm expansion on inequality. To this end, match key features of the US economy in 1980

- Define a region as 1990 labor market area, get $N = 391$ regions
- Enrich the model by adding:
 - Amenities B_i that enter utility multiplicatively
 - Local housing on which agents spend a fraction δ of their income. Constant housing supply elasticity ρ
- Divide the model's parameters into three groups:
 - Household preferences and housing market  Calibrate
 - Regional productivity and amenity fundamentals  Invert the model
 - Parameters from the production block  SMM

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Calibrating the household block

Parameter	Interpretation	Source	Value
ξ	Dispersion of location preference shocks	Galle et al. (2017)	2.8
ϵ	Dispersion of employer preference shocks	Lamadon et al. (2022), Berger et al. (2022)	5.0
σ	EoS between varieties	Costinot and Rodríguez-Clare (2014)	5.0
β_k	Sectoral expenditure shares	Direct computation - BEA NIPA (1980)	[0.64,0.36]
δ	Housing expenditure shares	Davis and Ortalo-Magne (2011)	0.24
ϱ	Housing supply elasticity	Saiz (2010)	1.75

- Also from a manual calculation using BEA KLEMS $\bar{\alpha}_s \equiv \{\alpha_s, \rho_s, \alpha_{g,s}\} = \{0.26, 0.36, 0.2\}$

Model inversion

- Need to recover $\mathcal{A} \equiv \left\{ \{A_{ij}\}_{j=1}^N, B_i, A_{g,i} \right\}_{i=1}^N$
- To recover A_{ij} , assume

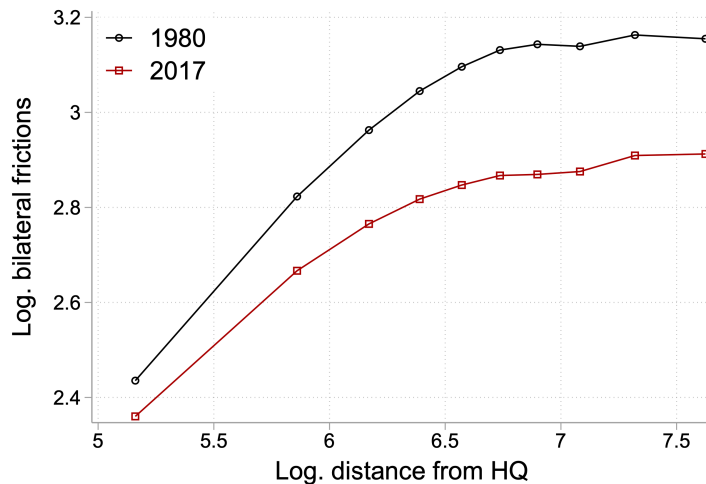
$$A_{ij} = A_i / \tau_{ij}$$

such that $\tau_{ij} > 1$ for $i \neq j$ and $\tau_{ii} = 1$.

- The set of τ_{ij} captures frictions for operating branches away from the HQ market
- Let M_{ij} be the measure of firms with headquarters in location i and a branch in location j . τ_{ij} can be shown to be equal

$$\tau_{ij} = \left(\frac{M_{ij} M_{ji}}{M_{ii} M_{jj}} \right)^{-\frac{1}{2\theta} \frac{\sigma-1}{\sigma}}$$

(a) Firm communication frictions against distance



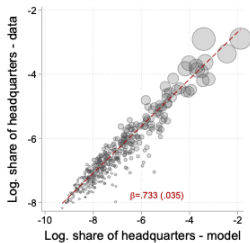
SMM for the production block

Minimize the loss function $\mathcal{L}(\Theta) \equiv (m(\Theta) - \tilde{m}_{1980})' W (m(\Theta) - \tilde{m}_{1980})$

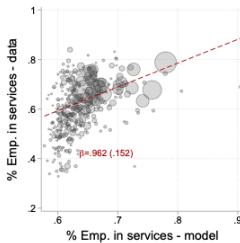
Parameter	Interpretation	Estimate	Main targeted moment	Model	Data (1980)
θ	Shape parameter of branch-level shocks	1.3	Var. of log. wages, MR firms - overall	0.22	0.22
η	Shape parameter of firm-level shocks	5.53	Var. of log. wages, MR firms - between firms	0.12	0.11
γ	Elasticity of branch productivity to HQ inputs	0.13	Var. of log. wages, MR firms - between HQ markets	0.05	0.05
c	Productivity loss from moving HQ	0.18	Regional covariance of headquarters and wages	0.16	0.2
f	Fixed cost of operating a branch	8.31	Average number of markets per multi-region firm	5.04	5.02
f_e	Fixed entry cost	301	Average employment per multi-region firm	462	484

Characterizing the baseline equilibrium

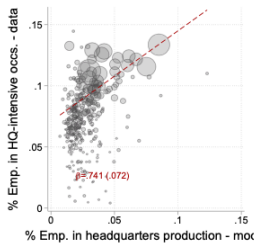
(a) Share of HQs



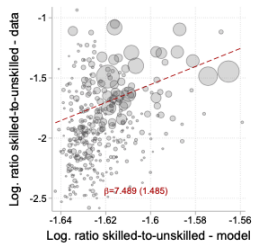
(b) % Emp. in services



(c) % Emp. in HQ prod.

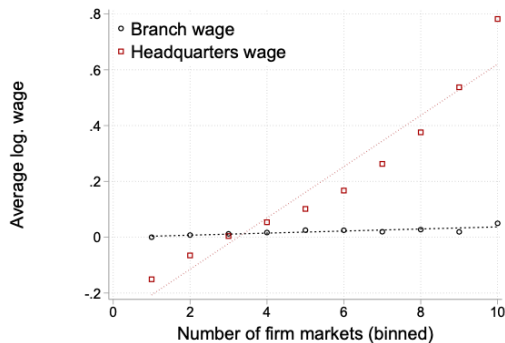


(d) Log. skill ratio

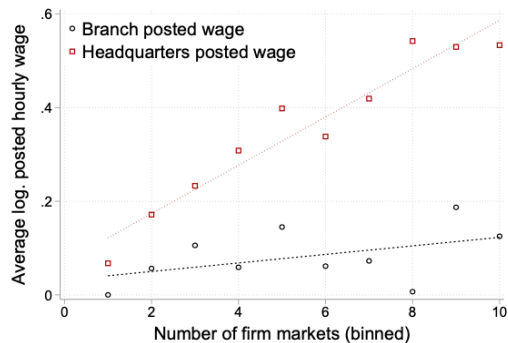


Characterizing the baseline equilibrium

(a) Within firm inequality - model



(b) Within firm inequality - data



Counterfactual analysis

Use the model to evaluate how shocks to the 1980 baseline equilibrium affect firm structure and inequality.

Focus on three important changes that were observed in the data:

- Reduction in frictions to firms expansion across space: lower τ_{ij}
- Change in preferences: higher expenditure on services: β from 64 to 79%
- Shock to reflect skill-biased tech change: increase in skill intensity of the educated

The question: if we make these changes to the 1980 baseline, how much does the new equilibrium resemble the 2017 one?

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Counterfactual analysis

	(1)	(2)	(3)	(4)	(5)	(6)	(7)
	Changes in levels						
	Data: $\Delta 1980-2017$	$\Delta \tau$	$\Delta \beta$	$\Delta(\tau, \beta)$	$\Delta \alpha$	$\Delta(\tau, \alpha)$	$\Delta(\tau, \beta, \alpha)$
	Cross-section of firms						
Log number of markets per firm	0.246	0.300	0.038	0.182	0.029	0.298	0.181
Variance of log wages: multi-region service firms	0.140	0.025	0.008	0.055	0.023	0.046	0.074
Variance of log wages: within-firms	0.070	0.015	0.021	0.045	0.031	0.035	0.055
	Cross-section of labor market areas						
Variance of log regional wages	0.034	0.007	0.003	0.022	0.002	0.012	0.038
Regression slope: log wage on log size	0.060	0.005	-0.016	0.011	0.005	0.017	0.025
Regression slope: log skill-premium on log size	0.052	0.011	0.009	0.012	0.003	0.023	0.025
Regression slope: log skill-ratio on log size	0.051	0.014	0.010	0.013	0.005	0.028	0.029

In percentages

More on the effect of small changes in τ

The spatial expansion of service firms in recent decades can help explain trends in US labor markets. This paper:

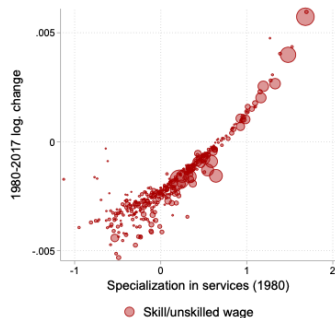
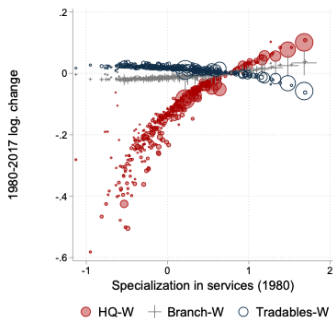
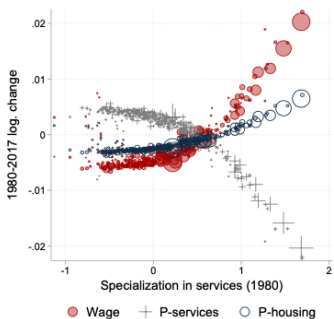
- Provides evidence on the uneven nature of firm expansion: firms expand more and skill concentration changes
- Develops a spatial general equilibrium model which provides predictions in line with the evidence
- Quantifies the model and suggests how it can account for a substantial share of the observed rise in inequality

Thank you!

More on the effects of τ

Remember we defined a measure of specialization in HQ services as $\frac{R_i}{E_i}$

Apply a small reduction in τ to the model and plot key regional variables against it:



Firm spatial expansion

Type of firm	Year	Firms	% of workforce	Emp. per firm	Estabs. per firm	CZs per firm
Goods-producing firms	1980	639800	31%	38.0	1.2	1.1
Goods-producing firms	2017	778400	14%	23.3	1.1	1.1
Services, single estab.	1980	2407000	40%	12.7	1.0	1.0
Services, single estab.	2017	3769000	38%	13.1	1.0	1.0
Services, multi-estab. and single-CZ	1980	69000	5%	53.2	2.9	1.0
Services, multi-estab. and single-CZ	2017	72500	7%	125.3	3.5	1.0
Services, multi-estab. and multi-CZ	1980	43500	24%	432.6	12.2	5.0
Services, multi-estab. and multi-CZ	2017	72500	41%	748.2	20.4	6.4

The headquarters wage premium [Back](#)

	(1)	(2)	(3)	(4)	(5)	(6)
Outcome: log posted wage	Single-year			Multi-year		
Occupational HQ-intensity	0.249*** (0.0753)	0.437*** (0.0380)	0.228*** (0.0350)	0.362*** (0.0713)	0.455*** (0.0357)	0.315*** (0.0323)
College requirement	0.406*** (0.0105)	0.348*** (0.00768)	0.347*** (0.00774)	0.387*** (0.0108)	0.335*** (0.00783)	0.335*** (0.00782)
HQ-intensity $\times$ Log CZs			0.0730*** (0.0239)			0.0129* (0.00743)
Log CZs						0.0503** (0.0216)
Commuting zone FE	Y	Y	Y	Y	Y	Y
Industry FE	Y	N	N	Y	N	N
Firm FE	N	Y	Y	N	Y	Y
Year FE	N	N	N	Y	Y	Y
Observations	3281863	3281863	3281863	7253996	7253996	7253996

Note: This table shows the results from a regression of a log posted wages in the Burning Glass data on various job characteristics. Columns (1)-(3) consider only 2019 - the year with the largest number of postings that include wages. Columns (4)-(6) consider a multi-year setting covering 2015-2019. Earlier years (2010-2014) are omitted due to limited information on posted wages. “Occupational HQ-intensity” captures the measure developed in Section 2.3.2. “College requirement” is an indicator that equals to one if a posting explicitly requires a college degree. “Log CZs” is the average number of commuting zones in which the firm is active. Standard errors in parentheses, clustered at the firm level. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

The industry composition of multi-region service firms

Firm activity	(1) % of firm payroll	(2) % of HQ-CZ payroll	(3) % of non-HQ-CZ payroll
Establishments in the firm's core sector	84.4%	67.7%	92.1%
Headquarters activity	11.2%	29.2%	3.0%
Other establishments	4.4%	3.1%	4.9%
Total	100%	100%	100%

Details on decomposition

Consider a partition of the universe of establishments into G distinct groups. The total change in variance equals to the sum of several components:

$$\begin{aligned}
 \underbrace{\Delta\sigma_t^2}_{\text{Change in variance}} &= \underbrace{s_{g',0} (\Delta\sigma_{g',t}^2)}_{\text{Variance within group}} + \underbrace{\sum_{g \in G} (\Delta s_{g,t}) \sigma_{g,0}^2}_{\text{Emp. reallocation across groups}} \\
 &+ \underbrace{\sum_{g \in G} (\Delta s_{g,t}) (\Delta\sigma_{g,t}^2)}_{\text{Comovement of variance and emp.}} + \underbrace{\sum_{g \in G/g'} s_{g,0} (\Delta\sigma_{g,t}^2)}_{\text{Residual}} + \underbrace{\sum_{g \in G} \Delta s_{g,t} (\mu_{g,t}^2 - \mu_t^2)}_{\text{Variance within other groups}}
 \end{aligned}$$

We can further decompose the increase in the variance for multi-region service firms, $\Delta\sigma_{\text{MR-serv},t}^2$, to changes within firms and changes between firms:

$$\Delta\sigma_{\text{MR-serv},t}^2 = \underbrace{\Delta_t \mathbb{V} [\ln \bar{w}_f]}_{\text{Between-firms - 55\%}} + \underbrace{\Delta_t \mathbb{V} [\ln w_{fj} - \ln \bar{w}_f]}_{\text{Within-firms, between establishments - 45\%}}$$

Details on decomposition: estimate

% of the overall increase in variance (1980-2017)	(1) Multi- estab. firms	(2) Service firms	(3) Multi-CZ services
Rising variance within the group of firms	83%	70%	40%
Changes in employment shares b/w groups (reallocation)	-4%	6%	8%
Comovement of variance and employment shares	9%	10%	20%
Residual	12%	13%	32%
Total change across all firms in the economy	100%	100%	100%

Solving the branch problem

Conditional on $h_i \equiv \prod_{s=1}^S h_{is}^{\rho_s}$ and on activity in market j , solve

$$\pi_{0,ij}(z) = \max_{\{l_s\}_{s=1}^S} E_j^{\frac{1}{\sigma}} P_j^{\frac{\sigma-1}{\sigma}} (q_{ij}(z))^{\frac{\sigma-1}{\sigma}} - \sum_{s=1}^S W_{js} L_{js}^{-\frac{1}{\epsilon}} l_{ijs}^{\frac{\epsilon+1}{\epsilon}}.$$

From which

$$r_{ij}(z_i, \phi_j) = (1 - \psi)^{\frac{1}{\psi}(1-\psi)} \left(E_j^{\frac{1}{\sigma}} P_j^{\frac{\sigma-1}{\sigma}} \right)^{\frac{1}{\psi}} C_j^{-\frac{1-\psi}{\psi}} \varphi_j(z_i, \phi_j)^{\frac{1}{\psi} \frac{\sigma-1}{\sigma}},$$

$$l_{ijs}(z_i, \phi_j) = W_{js}^{-\frac{\epsilon}{\epsilon+1}} L_{js}^{\frac{1}{\epsilon+1}} \left((1 - \psi) \alpha_s r_{ij}(z_i, \phi_j) \right)^{\frac{\epsilon}{\epsilon+1}}$$

$$w_{ijs}(z_i, \phi_j) = W_{js}^{\frac{\epsilon}{\epsilon+1}} L_{js}^{-\frac{1}{\epsilon+1}} \left((1 - \psi) \alpha_s r_{ij}(z_i, \phi_j) \right)^{\frac{1}{\epsilon+1}}$$

where ψ captures the share of operating profits from sales in market

$$\psi \equiv 1 - \frac{\sigma - 1}{\sigma} \frac{\epsilon}{1 + \epsilon}$$

C_j is a regional cost-shifter that summarizes the price and availability of labor for branch level production in that market $C_j \equiv \prod_s \left(\alpha_s W_{js}^{-1} L_{js}^{\frac{1}{\epsilon}} \right)^{-\alpha_s}$ and ϕ_j is the firm's productivity at market j

Solving the branch activity decision

The firm serves market j if these operating profits are greater than the fixed cost $\bar{P}_j f$. This decision takes place when the realization of the market-specific idiosyncratic shock ϕ_j is large enough, or $\psi r_{ij}(z) > \bar{P}_j f$.

$$\phi_j^{\frac{1}{\psi} \frac{\sigma-1}{\sigma}} > \frac{\bar{P}_j f}{\psi(1-\psi)^{\frac{1}{1-\psi}} \left(E_j^{\frac{1}{\sigma}} P_j^{\frac{\sigma-1}{\sigma}}\right)^{\frac{1}{\psi}} C_j^{-\frac{1-\psi}{\psi}} (A_{ij} \varphi_i(z))^{\frac{1}{\psi} \frac{\sigma-1}{\sigma}}}.$$

For convenience, denote $r_{0,ij}(z) \equiv (1-\psi)^{\frac{1}{\psi(1-\psi)}} \left(E_j^{\frac{1}{\sigma}} P_j^{\frac{\sigma-1}{\sigma}}\right)^{\frac{1}{\psi}} C_j^{-\frac{1-\psi}{\psi}} (A_{ij} \varphi_i(z))^{\frac{1}{\psi} \frac{\sigma-1}{\sigma}}$. The threshold productivity for activity is thus equal to

$$\tilde{\phi}_j^{\frac{1}{\psi} \frac{\sigma-1}{\sigma}} = \frac{\bar{P}_j f}{\psi r_{0,ij}(z)},$$

and the probability for activity is

$$\Pr \left(\phi_j^{\frac{1}{\psi} \frac{\sigma-1}{\sigma}} > \frac{\bar{P}_j f}{\psi r_{0,ij}(z)} \right) = \left(\frac{\bar{P}_j f}{\psi r_{0,ij}(z)} \right)^{-\theta}.$$

The resulting unconditional sales are

$$\bar{r}_{ij}(z) = \frac{\theta}{\theta - 1} \left(\frac{\bar{p}_j f}{\psi} \right)^{1-\theta} (r_{0,ij}(z))^\theta ,$$

which can also be written as

$$\bar{r}_{ij}(z) = \tilde{\psi} f^{1-\theta} \Upsilon_j (A_{ij} z_i h_i^\gamma)^{\frac{\theta}{\psi} \frac{\sigma-1}{\sigma}} ,$$

where

$$\tilde{\psi} \equiv \left(\frac{\theta}{\theta - 1} \psi^{\theta-1} (1 - \psi)^{\frac{\theta}{\psi} (1-\psi)} \right) ,$$

and the regional aggregate Υ_j captures market potential in location j

$$\Upsilon_j \equiv \left(E_j^{\frac{1}{\sigma}} P_j^{\frac{\sigma-1}{\sigma}} \bar{p}_j^{-\psi \frac{\theta-1}{\theta}} C_j^{-(1-\psi)} \right)^{\frac{\theta}{\psi}}$$

Headquarters and firm-wide solutions

Based on its expected revenues across all markets, the firm solves:

$$\max_{\{h_s\}_{s=1}^S} \tilde{\psi} f^{1-\theta} \left(\sum_{j=1}^N A_{ij}^{\frac{\theta}{\psi} \frac{\sigma-1}{\sigma}} \gamma_j \right) \left(z \left(\prod_s h_{is}^{\rho_s} \right)^\gamma \right)^{\frac{\theta}{\psi} \frac{\sigma-1}{\sigma}} - \sum_{s=1}^S W_{is} L_{is}^{-\frac{1}{\epsilon}} h_{is}^{\frac{\epsilon+1}{\epsilon}}.$$

The first order conditions yield

$$\underbrace{\rho_s \gamma \frac{\epsilon}{\epsilon+1} \frac{\sigma-1}{\sigma} \frac{\theta}{\psi}}_{1-\iota} \frac{\psi}{\theta} \bar{r}_i(z) = W_{is} L_{is}^{-\frac{1}{\epsilon}} h_{is}^{\frac{\epsilon+1}{\epsilon}}$$

and the resulting firm choices are

$$h_{is}(z) = \left(\rho_s (1-\iota) \frac{\psi}{\theta} W_{is}^{-1} L_{is}^{\frac{1}{\epsilon}} \bar{r}_i(z) \right)^{\frac{\epsilon}{\epsilon+1}}$$
$$W_{is}(z) = W_{is}^{\frac{\epsilon}{\epsilon+1}} L_{is}^{-\frac{1}{\epsilon+1}} \left(\rho_s (1-\iota) \frac{\psi}{\theta} \bar{r}_i(z) \right)^{\frac{1}{\epsilon+1}}.$$

where $\bar{r}_i(z_i)$ captures expected firm revenues across all markets.

Headquarters and firm-wide solutions

The expected total revenues $\bar{r}_i(z_i)$ take the simple form $\bar{r}_i(z_i) = \Lambda_i z_i^{\frac{\sigma-1}{\sigma} \frac{\theta}{\iota\psi}}$. where ι captures the share of operating profits that goes to owners, and Λ_i captures all forces that make location i more attractive for operating HQs:

$$\Lambda_i \equiv \underbrace{\tilde{t} \left(\sum_{j=1}^N A_{ij}^{\frac{\theta}{\psi} \frac{\sigma-1}{\sigma}} \Upsilon_j \right)^{\frac{1}{\iota}}}_{\text{HQ market access}} \underbrace{\left(\prod_s \left(\rho_s W_{is}^{-1} L_{is}^{\frac{1}{e}} \right)^{-\rho_s} \right)^{-\frac{1-\iota}{\iota}}}_{\text{Cost and supply of HQ labor}}$$

where $\tilde{t}$ is a constant that subsumes various parameters, and we define the headquarters labor cost shifter

$$\Upsilon_i \equiv \prod_s \left(\rho_s W_{is}^{-1} L_{is}^{\frac{1}{e}} \right)^{-\rho_s}.$$

Expected firm sales to specific market j are thus given by

$$\bar{r}_{ij}(z) = \frac{A_{ij}^{\frac{\theta}{\psi} \frac{\sigma-1}{\sigma}} \Upsilon_j}{\sum_{j'=1}^N A_{ij'}^{\frac{\theta}{\psi} \frac{\sigma-1}{\sigma}} \Upsilon_{j'}} \Lambda_i z^{\frac{\sigma-1}{\sigma} \frac{\theta}{\omega\psi}}$$

Entry and Headquarters locations

Can write the profits as

$$\mathbb{E}_\phi \left[\pi_i \left(c^{1^{i \neq n}} z_{0,i}, \phi \right) \right] = \frac{\psi \iota}{\theta} \Lambda_i z_i^{\frac{\sigma-1}{\sigma}} \frac{\theta}{\alpha \psi}$$

which is just a constant share $\psi \iota / \theta$ of expected revenue.

- ψ and ι contain σ and ϵ , capturing firm's market power in output and labor
- θ that governs the probability of serving individual markets
- ι governs the ability of the firm to raise revenues across all markets through the HQ inputs.

Log linearity and Frechet assumption give the probability that a firm that entered in location n chooses i as its headquarters base, denoted by λ_{ni} :

$$\lambda_{ni} = \frac{c^{\eta \mathbb{I}_{n \neq i}} \Lambda_i^{\eta / \left(\frac{\sigma-1}{\sigma} \frac{\theta}{\omega \psi} \right)}}{\sum_{m=1}^N c^{\eta \mathbb{I}_{n \neq m}} \Lambda_m^{\eta / \left(\frac{\sigma-1}{\sigma} \frac{\theta}{\alpha \psi} \right)}}.$$

And see that if $c \rightarrow 0$ all firms stay in the entry market and there is no sorting based on the set of productivity draws $\{z_{0,i}\}_{i=1}^N$. In this case, $\lambda_{ni} = 1$ for $n = i$ and is zero otherwise.

Entry and Headquarters locations

Ex ante expected profits for market n are given by

$$\pi_{e,n} = \tilde{\pi} \left[(1 - c^\eta) \Lambda_n^{\eta / \left(\frac{\sigma-1}{\sigma} \frac{\theta}{\omega \psi} \right)} + \sum_{i=1}^N c^\eta \Lambda_i^{\eta / \left(\frac{\sigma-1}{\sigma} \frac{\theta}{\omega \psi} \right)} \right]^{\left(\frac{\sigma-1}{\sigma} \frac{\theta}{\omega \psi} \right) / \eta}$$

with $\tilde{\pi} \equiv \frac{\psi_\iota}{\theta} \Gamma \left(1 - \left(\frac{\sigma-1}{\sigma} \frac{\theta}{\omega \psi} \right) / \eta \right)$ where $\Gamma(\cdot)$ is the Gamma-function. This expression shows that ex-ante profits depend on the HQ-attractiveness shifters Λ_i across all locations given the ability to place headquarters away from the firm's entry market n .

- $c \rightarrow 0$, expected profits simply scale with the value of Λ_n in the entry location n .
- $c \rightarrow 1$ expected profits become a standard power mean of Λ_i across all locations, and entry cost is sunk.

Characterizing firm wages

From the branch problem FOCs:

$$w_{ijs}(z, \phi) = W_{js}^{\frac{\epsilon}{\epsilon+1}} L_{js}^{-\frac{1}{\epsilon+1}} \left((1-\psi)\alpha_s(1-\psi)^{\frac{1}{\psi}(1-\psi)} \left(E_j^{\frac{1}{\sigma}} P_j^{\frac{\sigma-1}{\sigma}} \right)^{\frac{1}{\psi}} C_j^{-\frac{1-\psi}{\psi}} \varphi_{ij}(z, \phi)^{\frac{1}{\psi} \frac{\sigma-1}{\sigma}} \right)^{\frac{1}{\epsilon+1}}.$$

Using HQ FOC, $\varphi_{ij}(z, \phi) = A_{ij} z \phi \left(\prod_{s=1}^S h_{is}^{\rho_s} \right)^{\gamma} = A_{ij} \phi \left((1-\iota)^{\frac{\psi}{\theta}} \frac{\Lambda_i}{\Gamma_i} \right)^{\frac{\epsilon}{\epsilon+1} \gamma} z^{\frac{1}{\iota}}$. Plugging it into the expression for branch wages yields

$$w_{ijs}(z, \phi) = \text{const}_s W_{js}^{\frac{\epsilon}{\epsilon+1}} L_{js}^{-\frac{1}{\epsilon+1}} \left(\left(E_j^{\frac{1}{\sigma}} P_j^{\frac{\sigma-1}{\sigma}} \right)^{\frac{1}{\psi}} C_j^{-\frac{1-\psi}{\psi}} \right)^{\frac{1}{\epsilon+1}} \\ \times \left(\left(A_{ij} \phi \left(\frac{\Lambda_i}{\Gamma_i} \right)^{\frac{\epsilon}{\epsilon+1} \gamma} z^{\frac{1}{\iota}} \right)^{\frac{1}{\psi} \frac{\sigma-1}{\sigma}} \right)^{\frac{1}{\epsilon+1}}$$

Substituting $\left(E_j^{\frac{1}{\sigma}} P_j^{\frac{\sigma-1}{\sigma}} C_j^{-(1-\psi)} \right)^{\frac{1}{\psi}} = \Upsilon_j^{\frac{1}{\theta}} \bar{P}_j^{\frac{1}{\theta}-1}$ from the definition of the market potential term Υ_j gives the expression. [Back](#)

Characterizing firm wages

HQ wages are given by

$$w_{is}(z) = W_{is}^{\frac{\epsilon}{\epsilon+1}} L_{is}^{-\frac{1}{\epsilon+1}} \left(\rho_s (1 - \iota) \frac{\psi}{\theta} \Lambda_i z^{\frac{\sigma-1}{\sigma} \frac{\theta}{\iota \psi}} \right)^{\frac{1}{\epsilon+1}}.$$

Now consider two i -headquartered firms with firm productivities z and z' that are active in market j and have the same branch level productivity ϕ . Using the branch level wage expression we get

$$\left(\frac{w_{is}(z)}{w_{ijs}(z, \phi)} / \frac{w_{is}(z')}{w_{ijs}(z', \phi)} \right) = \left(\frac{z}{z'} \right)^{(\theta-1) \frac{\sigma-1}{\sigma} \frac{1}{\iota \psi} \frac{1}{\epsilon+1}}.$$

Since $\theta > 1$, this ratio is increasing in z/z' . Therefore, the HQ-branch wage differential is higher in the more productive firm, yielding the result.

Inequality for a given firm type

- The ratio

$$\frac{\text{total expected firm revenues across all locations}}{\text{to local revenues in the headquarters market}} = \frac{r_i(z_i)}{r_{ii}(z_i, \phi_i)}$$

can be used as measure of firm expansion in space.

- Let $r_{ij}(z_i, \phi_j)$ be firm revenues in some market j . For each skill type s ,

$$\frac{w_{is}^{HQ}(z_i)}{w_{iis}(z_i, \phi_i)} \text{ is increasing in } \frac{r_i(z_i)}{r_{ii}(z_i, \phi_i)}$$

Aggregation of firm revenues

Aggregating expected firm sales, one gets

$$R_{ij} = \frac{A_{ij}^{\frac{\theta}{\psi} \frac{\sigma-1}{\sigma}} \Upsilon_j}{\sum_{j=1}^N A_{ij'}^{\frac{\theta}{\psi} \frac{\sigma-1}{\sigma}} \Upsilon_{j'}} \Lambda_i Z_i M_i$$

where M_i is the mass of firms with headquarters in market i , and Z_i is an aggregate term capturing the average productivity of firms with headquarters in location i .
Total sales by firms headquartered in location i are thus given by

$$R_i = \sum_{j=1}^N R_{ij}$$

Defining $\chi_{ji} = \frac{A_{ij}^{\frac{\theta}{\psi} \frac{\sigma-1}{\sigma}} \left(\frac{\Lambda_i}{\Gamma_i}\right)^{1-\iota} Z_i M_i}{\sum_{n=1}^N A_{nj}^{\frac{\theta}{\psi} \frac{\sigma-1}{\sigma}} \left(\frac{\Lambda_n}{\Gamma_n}\right)^{1-\iota} Z_n M_n}$ we can write compactly $R_{ij} = \chi_{ji} E_j$

Services price index

To get the services price index, first note that

$$E_j = \sum_i R_{ij} = \Upsilon_j \sum_i \tilde{\psi} f^{1-\theta} A_{ij}^{\frac{\theta}{\psi} \frac{\sigma-1}{\sigma}} \left((1-\iota) \frac{\psi}{\theta} \frac{\Lambda_i}{C_i} \right)^{1-\iota} Z_i M_i.$$

Then from the definition of Υ_i , with some algebra

$$\begin{aligned} P_i^{-\frac{\theta}{\psi} \frac{\sigma-1}{\sigma}} &= \bar{P}_i^{1-\theta} E_i^{\frac{\theta}{\psi} \frac{1}{\sigma}} C_i^{-\frac{\theta}{\psi} (1-\psi)} \Upsilon_i^{-1} \\ &= \tilde{\psi} \left((1-\iota) \frac{\psi}{\theta} \right)^{1-\iota} \bar{P}_i^{1-\theta} E_i^{\frac{\theta}{\psi \sigma} - 1} C_i^{-\frac{\theta}{\psi} (1-\psi)} \sum_n f^{1-\theta} A_{ni}^{\frac{\theta}{\psi} \frac{\sigma-1}{\sigma}} \left(\frac{R_n}{\Gamma_n} \right)^{1-\iota} (M_n Z_n)^\iota \end{aligned}$$

So that

$$P_i = \left[\tilde{\psi} \bar{P}_i^{1-\theta} E_i^{\frac{\theta}{\psi \sigma} - 1} C_i^{-\frac{\theta}{\psi} (1-\psi)} \sum_n A_{ni}^{\frac{\theta}{\psi} \frac{\sigma-1}{\sigma}} \left(\frac{R_n}{\Gamma_n} \right)^{1-\iota} (M_n Z_n)^\iota \right]^{\frac{1}{-\frac{\theta}{\psi} \frac{\sigma-1}{\sigma}}}$$

Equilibrium in regional aggregates

Labor market clearing determines the wage levels W_{is} and employment levels L_{is} . This block includes the wage-index the sorting of households across space:

$$\frac{L_{is}}{\sum_{j=1}^N L_{js}} = \frac{(W_{is}/\bar{P}_i)^\xi}{\sum_{j=1}^N (W_{js}/\bar{P}_j)^\xi}$$

Services market clearing and regional balance of payments: Services price indices P_i , expenditure levels E_i , and revenues R_i , are determined by market clearing in the services market,

$$R_i = \sum_{j=1}^N \frac{A_{ij}^{\frac{\theta}{\psi} \frac{\sigma-1}{\sigma}} \left(\frac{\Lambda_j}{\Gamma_i} \right)^{1-\iota} Z_j M_j}{\sum_{n=1}^N A_{nj}^{\frac{\theta}{\psi} \frac{\sigma-1}{\sigma}} \left(\frac{\Lambda_n}{\Gamma_n} \right)^{1-\iota} Z_n M_n} E_j$$

the expression for the services price index

$$P_i = \left[\tilde{c} \bar{P}_i^{1-\theta} E_i^{\frac{\theta}{\psi\sigma}-1} C_i^{-\frac{\theta}{\psi}(1-\psi)} \sum_n A_{ni}^{\frac{\theta}{\psi} \frac{\sigma-1}{\sigma}} \left(\frac{R_n}{\Gamma_n} \right)^{1-\iota} (M_n Z_n)^\iota \right]^{-\frac{1}{\psi \frac{\theta-1}{\sigma}}}$$

and the regional balance of payments,

$$\left(\frac{1-\beta}{\beta} + \frac{\psi}{\theta} \right) E_i = R_{g,i} + (1-\iota) \frac{\psi}{\theta} R_i + M_{e,i} \bar{P}_i f_e,$$

with

$$\Lambda_i \equiv \tilde{\iota} \sum_{j=1}^N A_{ij}^{\frac{\theta}{\psi} \frac{\sigma-1}{\sigma}} \Upsilon_j^{\frac{1}{\psi}} \Gamma_i^{-\frac{1-\iota}{\psi}}, \quad \Gamma_i \equiv \prod_s \left(\rho_s W_{is}^{-1} L_{is}^{\frac{1}{\epsilon}} \right)^{-\rho_s}$$

and

$$\Upsilon_j \equiv \left(E_j^{\frac{1}{\sigma}} P_j^{\frac{\sigma-1}{\sigma} - \beta \psi \frac{\theta-1}{\theta}} C_j^{-(1-\psi)} \right)^{\frac{\theta}{\psi}}, \quad C_j \equiv \prod_s \left(\alpha_s W_{js}^{-1} L_{js}^{\frac{1}{\epsilon}} \right)^{-\alpha_s}$$

and that the aggregate price index $\bar{P}_i$ in the equilibrium with housing is given by

$$\bar{P}_i = \left(\frac{\delta}{\beta(1-\delta)} \right)^{\frac{\delta}{(1+\varrho)-\delta\varrho}} P_i^{\beta(1-\frac{\delta}{(1+\varrho)-\delta\varrho})} E_i^{\frac{\delta}{(1+\varrho)-\delta\varrho}}$$

Equilibrium in regional aggregates

Tradable goods. Tregional mass of tradable-goods firms $M_{g,i}$ and revenues $R_{g,i}$ are determined by the optimality and free entry conditions for this sector:

$$R_{g,i} = M_{g,i} \left(\left(\frac{\epsilon}{1+\epsilon} \frac{1}{C_{gi}} \right)^{\frac{\epsilon}{1+\epsilon}} A_{g,i} \right)^{1+\epsilon}, \quad C_{g,i} \equiv \prod_s \left(\alpha_{g,s} W_{is}^{-1} L_{is}^{\frac{1}{\epsilon}} \right)^{-\alpha_{g,s}}$$

and

$$M_{g,i} \bar{P}_i f_{g,i} = \frac{1}{1+\epsilon} R_{g,i}$$

Entry and optimal headquarters-location decisions in the services sector. All remaining aggregates $M_i, M_{e,i}, Z_i$ and $Z_{w,i}$ are determined in the entry and HQ-location block for the services sector. The aggregate productivity index Z_i is given by

$$Z_i = \Gamma \left(1 - 1/ \left(\frac{\eta}{\zeta} \right) \right) \sum_o \frac{M_{e,o}}{M_i} c^{\zeta \mathbb{I}_{o \neq i}} \lambda_{oi}^{1 - \frac{\zeta}{\eta}}$$

and the labor market productivity term $Z_{w,i}$ is given by

$$Z_{w,i} = \Gamma \left(1 - 1/ \left(\frac{\eta}{\zeta_w} \right) \right) \sum_o \frac{M_{e,o}}{M_i} \lambda_{oi}^{1 - \frac{\zeta}{\eta}} c^{\zeta_w \mathbb{I}_{o \neq i}},$$

where $\Gamma(\cdot)$ is the Gamma-function, ζ is a constant of parameters given by $\zeta \equiv \frac{\sigma-1}{\sigma} \frac{\theta}{\nu\psi}$, and the headquarters probability shares λ_{ni} are given by

$$\lambda_{ni} = \frac{(c^{\zeta \mathbb{I}_{n \neq i}} \Lambda_i)^{\frac{\eta}{\zeta}}}{\sum_{m=1}^N (c^{\zeta \mathbb{I}_{n \neq m}} \Lambda_m)^{\frac{\eta}{\zeta}}}.$$

The mass of entry $M_{e,i}$ for each market can be recovered from the condition that balances the flow of firms across locations

$$M_i = \sum_n \lambda_{ni} M_{e,n}$$

Even more on inversion: A_i

Market clearing in the services sector implies that the total sales of all i headquartered firms, R_i , must equal to the total spending on them across all locations

$$R_i = \sum_{j=1}^N \frac{(A_i/\tau_{ij})^{\frac{\theta}{\psi} \frac{\sigma-1}{\sigma}} \left(\frac{\Lambda_i}{\Gamma_i}\right)^{1-\iota} Z_i M_i}{\sum_{n=1}^N (A_n/\tau_{nj})^{\frac{\theta}{\psi} \frac{\sigma-1}{\sigma}} \left(\frac{\Lambda_n}{\Gamma_n}\right)^{1-\iota} Z_n M_n} E_j.$$

So with data on R_i across all locations recover a guess for A_i (up to scale) from the above set of equations, and iterate over it as part of the solution for equilibrium.

Even more on inversion: B_i

To get the set of regional amenities $\{B_i\}_{i=1}^N$, invert the equilibrium conditions for worker sorting across space. Recall that the probability that a worker of type s chooses region i is given by

$$\frac{L_{is}}{\bar{L}_s} = \frac{(B_i W_{is} / \bar{P}_i)^\xi}{\sum_{j=1}^N (B_j W_{js} / \bar{P}_j)^\xi},$$

where $\bar{L}_s$ is the aggregate exogenous measure of type- s workers in the economy. Multiplying by $\bar{L}_s$ and summing across all skill groups, we obtain an expression for the total equilibrium level of employment in each region,

$$L_i = \sum_{s=1}^S \frac{(B_i W_{is} / \bar{P}_i)^\xi}{\sum_{j=1}^N (B_j W_{js} / \bar{P}_j)^\xi} \bar{L}_s.$$

Using data on the observed allocation of employment across space, $\{L_i\}_{i=1}^N$, use the above expression to exactly recover $\{B_i\}_{i=1}^N$. Then, can recover the group-specific measures of workers from above.

Even more on inversion: $A_{g,i}$

To recover $\{A_{g,i}\}_{i=1}^N$, first note that the total regional wage bill in the model can be written as

$$L_i W_{a,i} = \underbrace{\frac{\epsilon}{\epsilon+1} \frac{\sigma-1}{\sigma} (E_i + \gamma R_i)}_{\text{Regional wage bill from services}} + \underbrace{\frac{\epsilon}{\epsilon+1} R_{g,i}}_{\text{Regional wage bill from tradable-goods}},$$

where $W_{a,i}$ is the employment-weighted arithmetic mean of wages in location i . Therefore, conditional on parameters and values for R_i and E_i , we recover the implied revenues in the tradable-goods sector $R_{g,i}$. Then, use the equilibrium condition for tradable-goods revenues

$$R_{g,i} = M_{g,i} \left(\left(\frac{\epsilon}{1+\epsilon} \frac{1}{C_{g,i}} \right)^{\frac{\epsilon}{1+\epsilon}} A_{g,i} \right)^{1+\epsilon}, \quad C_{g,i} \equiv \prod_s \left(\alpha_{g,s} W_{is}^{-1} L_{is}^{\frac{1}{\epsilon}} \right)^{-\alpha_{g,s}},$$

to recover the set of productivities $\{A_{g,i}\}_{i=1}^N$.

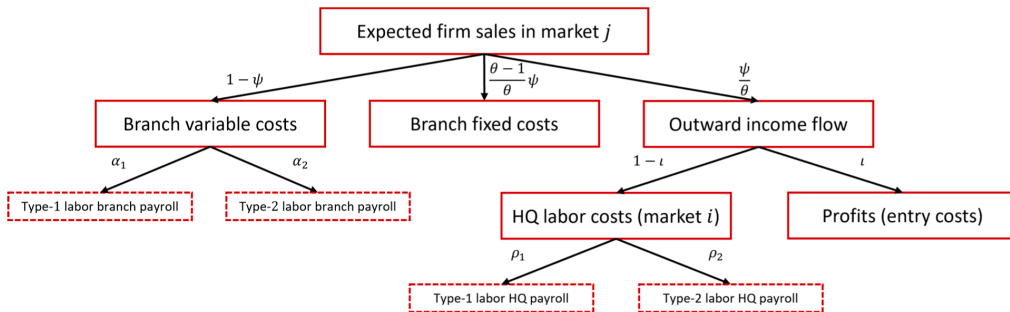
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Counterfactual analysis: percentage changes

	Data: $\Delta 1980-2017$	$\Delta \tau$	$\Delta \beta$	$\Delta(\tau, \beta)$	$\Delta \alpha$	$\Delta(\tau, \alpha)$	$\Delta(\tau, \beta, \alpha)$
	% of changes in data (1980-2017)						
	Cross-section of firms						
Log number of markets per firm	100%	122%	16%	74%	12%	121%	74%
Variance of log wages: multi-region service firms	100%	18%	6%	39%	17%	33%	53%
Variance of log wages: within-firms	100%	21%	30%	64%	44%	50%	79%
	Cross-section of labor market areas						
Variance of log regional wages	100%	21%	10%	65%	7%	35%	112%
Regression slope: log wage on log size	100%	8%	-26%	19%	8%	29%	41%
Regression slope: log skill-premium on log size	100%	22%	17%	23%	7%	44%	48%
Regression slope: log skill-ratio on log size	100%	28%	20%	26%	10%	56%	57%

Composition of firm revenues

Figure 21: The composition of firm revenues in the model



Note: This figure shows how the expected sales of a i -headquartered firm in some market j are distributed across locations and different types of costs. “Outward income flow” stands for the residual of local revenues after paying local variable and fixed costs. The figure considers two types of worker skills, such that $S = \{1, 2\}$. The composite parameters ψ and ι are given by $\psi \equiv 1 - \frac{\sigma-1}{\sigma} \frac{\epsilon}{1+\epsilon}$ and $\iota \equiv 1 - \gamma \frac{\epsilon}{\epsilon+1} \frac{\sigma-1}{\sigma} \frac{\theta}{\psi}$.

Skill biased technological change

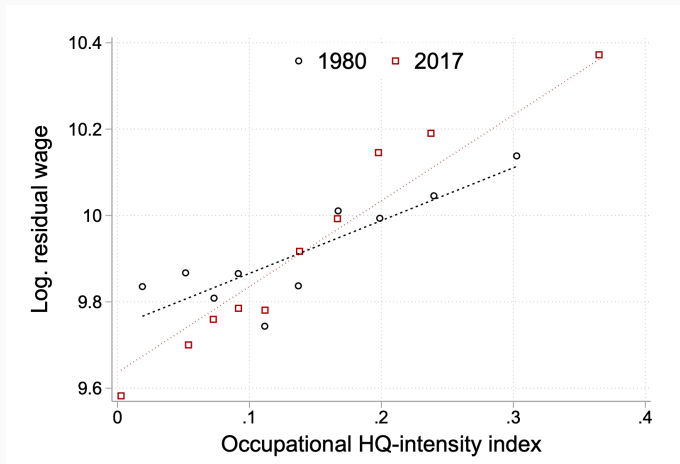
The model provides a set of predictions that go beyond changes in skill-premium:

- Conditional on worker skill and location, workers at headquarters-level jobs earn higher wages
- This relationship has a systematic connection to firm size, with greater HQ-branch gaps in larger firms
- This conditional headquarters wage premium has risen over time, given the increase in the geographical scope of firms.

For further evidence, the paper turns to the American Community Survey (ACS)

The conditional HQ premium

The figure plots the average earnings of workers against HQ intensity, after controlling for educational attainment, interacted with worker commuting zone.



Variance decomposition of individual earnings

To show how the HQ-branch differentials can impact inequality, perform a variance decomposition of individual earnings over time. Estimate the wage regression

$$\log w_i = \underbrace{\beta^{HQ} \text{HQ}_{occ(i)}}_{\text{Occupational HQ-intensity}} + \underbrace{\text{College}_i + \text{CZ}_i + \text{College}_i \times \text{CZ}_i}_{\text{College attainment and commuting zone}} + \epsilon_i,$$

- We have already seen β^{HQ} is positive and has increased over time
- The table shows that the increase in β^{HQ} can account for a large part of the increase in the wage inequality, even after controls.

Component of log individual earnings	Δ Variance 1980 – 2017	% of total Δ
Commuting zone and college attainment	0.03	18%
HQ-intensity of the occupation	0.08	46%
Covariance	0.03	20%
Residual	0.03	16%
Total	0.17	100%

More on inversion

Recover unilateral fundamentals $A_i, A_{g,i}, B_i$ by solving for the model's equilibrium with inversion of specific equilibrium conditions.

- The A_i are obtained by exactly matching the aggregate revenues of all i -headquartered firms, R_i .
 - These measures are obtained by attributing firm-level revenues for all service-sector firms in the Dun & Bradstreet data to their headquarters location, as implied by the model.
- $A_{g,i}$ is obtained as a residual to match exactly the aggregate wage bill in each LMA from the BLS Quarterly Census of Employment and Wages (QCEW).
- The set of amenities B_i is recovered by exactly matching the distribution of 1980 employment across regions.

Frame Title

	Data: $\Delta 1980-2017$	$\Delta \tau$	$\Delta \beta$	$\Delta(\tau, \beta)$	$\Delta \alpha$	$\Delta(\tau, \alpha)$	$\Delta(\tau, \beta, \alpha)$
	% of changes in data (1980-2017)						
	Cross-section of firms						
Log number of markets per firm	100%	122%	16%	74%	12%	121%	74%
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Equilibrium characterization: Aggregation

Lemma

The model's equilibrium can be expressed as a set of non-linear equations that include as variables only region-level aggregates.

Firm-level decisions are then obtained as power functions of these regional variables and the exogenous firm productivity terms z and ϕ .

The equilibrium regional variables:

- Regional employment and wage indices L_{is} and W_{is} , services price index P_i
- Mass of service entrants, service HQs and goods firms $M_{e,i}$, M_i and $M_{g,i}$
- Aggregate revenues for i -headquartered service firms and goods firms, R_i and $R_{g,i}$
- Average productivity indices Z_i , capturing the average HQ-level productivity in i .

Aggregate and regional skill-intensity

Suppose that s is the group of high-skilled workers. The economy-wide payroll-share of high-skilled workers is given by

$$\frac{\alpha_s + \rho_s \gamma + \alpha_{g,s} \frac{\sigma}{\sigma-1} \frac{1-\beta}{\beta}}{1 + \gamma + \frac{\sigma}{\sigma-1} \frac{1-\beta}{\beta}}$$

To get this, first note the total compensation for labor of type s in location i is given by

$$L_{is} W_{a,is} = \underbrace{\alpha_s \frac{\sigma-1}{\sigma} \frac{\epsilon}{\epsilon+1} E_i}_{\text{Branch-level compensation}} + \underbrace{\rho_s \gamma \frac{\epsilon}{\epsilon+1} \frac{\sigma-1}{\sigma} R_i}_{\text{HQ compensation}} + \underbrace{\alpha_{g,s} \frac{\epsilon}{1+\epsilon} R_{g,i}}_{\text{Tradable-goods compensation}},$$

where note that $W_{a,is}$ is the arithmetic mean regional wage for group s . To get the share of group s in total aggregate labor compensation, sum the above across all regions, and divide by the sum across all regions and skill groups, to obtain

$$\frac{W_{a,s} L_s}{W_a L} = \frac{\alpha_s + \rho_s \gamma + \alpha_{g,s} \frac{\sigma}{\sigma-1} \frac{1-\beta}{\beta}}{1 + \gamma + \frac{\sigma}{\sigma-1} \frac{1-\beta}{\beta}}$$

Formulas for main aggregates

Γ_i captures the cost of headquarters labor in i :

$$\Gamma_i \equiv \prod_s \left(\rho_s W_{is}^{-1} L_{is}^{\frac{1}{\epsilon}} \right)^{-\rho_s}.$$

Λ_i captures the attractiveness of operating headquarters in i :

$$\Lambda_i \equiv \underbrace{\tilde{t} \left(\sum_{j=1}^N A_{ij}^{\frac{\theta}{\psi} \frac{\sigma-1}{\sigma}} \Upsilon_j \right)^{\frac{1}{\epsilon}}}_{\text{HQ market access}} \underbrace{\left(\prod_s \left(\rho_s W_{is}^{-1} L_{is}^{\frac{1}{\epsilon}} \right)^{-\rho_s} \right)^{-\frac{1-\alpha}{t}}}_{\text{Cost and supply of HQ labor}}$$

Υ_j captures the market-potential of operating a branch in j

$$\Upsilon_j \equiv \left(E_j^{\frac{1}{\sigma}} P_j^{\frac{\sigma-1}{\sigma}} \bar{P}_j^{-\psi \frac{\theta-1}{\theta}} C_j^{-(1-\psi)} \right)^{\frac{\theta}{\psi}}$$

Details on inversion: recovering the A_{ij}

In the model, M_{ij} can be shown to equal

$$M_{ij} = \frac{\theta - 1}{\theta} \frac{\psi}{\bar{P}_{jf}} \frac{A_{ij}^{\frac{\theta}{\psi} \frac{\sigma-1}{\sigma}} \left(\frac{\Lambda_i}{\Gamma_i} \right)^{1-\iota} Z_i M_i}{\sum_{n=1}^N A_{nj}^{\frac{\theta}{\psi} \frac{\sigma-1}{\sigma}} \left(\frac{\Lambda_n}{\Gamma_n} \right)^{1-\iota} Z_n M_n} E_j.$$

- Multiplying by M_{ji} and dividing by $M_{ii}M_{jj}$ yields an expression that depends only on τ_{ij} and model parameters.
- This can then be inverted to get to

$$\tau_{ij} = \left(\frac{M_{ij}M_{ji}}{M_{ii}M_{jj}} \right)^{-\frac{1}{2\theta} \frac{\sigma-1}{\sigma}}$$